Exercise 4.1

By Laplace transformation, the following transfer function is obtained (from force to velocity)

$$G_v(s) = \frac{V(s)}{F(s)} = \frac{1}{ms+b} = \frac{1/b}{\frac{m}{b}s+1}$$

Since the position is controlled, the considered system is

$$G(s) = \frac{P(s)}{F(s)} = \frac{1/b}{s(m/b\,s+1)}$$

If a proportional controller ($K(s) = K_p$) is used, then the loop gain is

$$L(s) = K_p \frac{1/b}{s(m/b\,s+1)}$$

Hence, the system is of type 1. Therefore, one can use a <u>P-controller.</u>

The closed-loop dynamics is given by

$$H(s) = \frac{K_p G(s)}{1+K_p G(s)}$$

$$= \frac{K_p/b}{m/b\,s^2 + s + K_p/b}$$

$$= \frac{K_p/m}{s^2 + \frac{b}{m}s + K_p/m}$$

Now that the transfer function is on standard form, it is seen that

$$2\xi\omega_n = \frac{b}{m} \quad \text{and} \quad \omega_n^2 = \frac{K_p}{m}$$

This implies that

$$\xi = \frac{b}{2\sqrt{K_p m}}$$

A critically damped system has $\xi = 1$. In this case

$$\underline{K_p = \frac{b^2}{4m}}$$